FIITJEE

ALL INDIA TEST SERIES

PART TEST – I

JEE (Advanced)-2020

PAPER – 1

TEST DATE: 10-11-2019

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A

Sol. $mg(r - R\sin\theta) = MgR\sin\theta$

$mr = (M + m)R\sin\theta$

$$\sin\theta = \frac{mr}{(M+m)R} = \frac{1}{2}$$

$\therefore \theta = 30°$

2. B

Sol. $(N_1 + N_2)\cos 30° = mg\cos\theta$

$$N_1 + N_2 = \frac{2}{\sqrt{3}}mg\cos\theta \quad \ldots(i)$$

$(N_2 - N_1)\cos 60° = mg\sin\theta$

$N_2 - N_1 = 2mg\sin\theta$ …(ii)

Solving (i) and (ii), we get

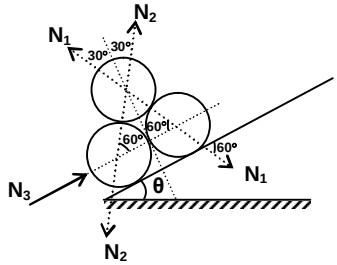
$$N_2 = \frac{mg}{\sqrt{3}}(\cos\theta + \sqrt{3}\sin\theta) \quad \ldots(iii)$$

$$N_1 = \frac{mg}{\sqrt{3}}(\cos\theta - \sqrt{3}\sin\theta) \quad \ldots(iv)$$

Also, $N_1\cos 60° = mg\sin\theta$

$N_1 = 2mg\sin\theta$ …(v)

From (iv) and (v),



$$2mg\sin\theta = \frac{mg}{\sqrt{3}}(\cos\theta - \sqrt{3}\sin\theta)$$

$$2\sqrt{3}\sin\theta = \cos\theta - \sqrt{3}\sin\theta$$

$$\Rightarrow 3\sqrt{3}\sin\theta = \cos\theta$$

$$\theta = \tan^{-1}\left(\frac{1}{3\sqrt{3}}\right)$$

3. A

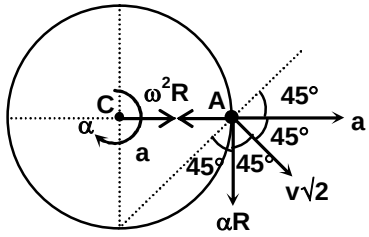
Sol. $a_{1(max)} = g(\mu\cos\theta - \sin\theta) = \frac{10}{13}$ m/s^2 (where $\mu = 0.5$ and $\mu_0 = 0.33$)

$F - (m_1 + m_2)\, g(\sin\theta + \mu_0\cos\theta) = (m_1 + m_2)\, a_{1(max)}$

$\Rightarrow F = 2000$ N

4. B

Sol. velocity of point 'A' $v_A = \sqrt{v^2 + \omega^2R^2} = v\sqrt{2}$



Normal acceleration of point A,

$a_A(n) = \omega^2R\cos 45° + \alpha R\cos 45° - a\cos 45°$

$a_{A(n)} = \frac{\omega^2R}{\sqrt{2}} = \frac{v^2}{\sqrt{2}R}$

$\therefore$ radius of curvature of trajectory of point 'A' relative to the ground is

$$r = \frac{(v_A)^2}{a_{A(n)}} = \frac{(v\sqrt{2})^2}{\frac{v^2}{\sqrt{2}R}} = 2\sqrt{2}R$$

5. A

Sol. $\theta = 30°$

6. A, C

Sol. $f_{s(max)} = \mu N = 2\tan\alpha \cdot W\cos\alpha = 2W\sin\alpha$

$H_{min} = \sqrt{(2W\sin\alpha)^2 - (W\sin\alpha)^2} = \sqrt{3}W\sin\alpha$

$H_{min} = \sqrt{3}W\sin\alpha$, $\phi = 30°$

7. A, C, D

Sol. The velocity parallel to the plane is unaltered by the impacts, so that the distance described parallel to the plane will be zero at the end of a time t given by :

$$0 = vt\cos(\theta - \alpha) - \frac{(g\sin\alpha)t^2}{2}$$

$$\Rightarrow \quad t = \frac{2v\cos(\theta - \alpha)}{g\sin\alpha}$$

Also, since the elasticity is perfect, the velocity perpendicular to the plane is just reversed at each impact. The time of flight for each trajectory is thus twice the time in which the velocity vsin $(\theta - \alpha)$ is destroyed by g cos α, and thus t = $\frac{2v\sin(\theta-\alpha)}{g\cos\alpha}$

Clearly the particle will return to the point of projection if the first of these is some multiple, n, of the second, i.e. if

$$\frac{2v\cos(\theta-\alpha)}{g\sin\alpha} = n\frac{2v\sin(\theta-\alpha)}{\cos\alpha}$$

i.e. if cot α . cot $(\theta - \alpha)$ is an integer.

8. C, D

Sol. $\frac{m}{M} = \frac{\sin(2\phi+\theta)}{\sin\theta}$

9. C, D

Sol. F_μ must be not less than $\frac{L(k^2+a^2)}{2a(2k^2+a^2)}$ =2 N

Let F be the friction on each of the front wheel in the forward direction, and F_1 the friction backward , on each of the hind wheel . Then :

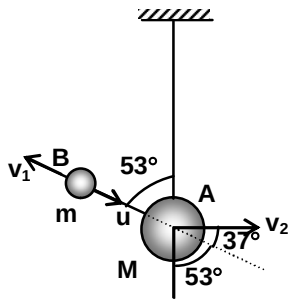
$$Mk^2\frac{d^2\theta}{dt^2} = L - 2Fa \;;\; Mk^2\frac{d^2\theta}{dt^2} = 2F_1a \text{ and } Ma\frac{d^2\theta}{dt^2} = 2F - 2F_1$$

Solving we get, $F = \frac{L(k^2+a^2)}{2a(2k^2+a^2)} = 2N$

F_μ must be not less than $\frac{L(k^2+a^2)}{2a(2k^2+a^2)}$ =2N

10. A, B, D

Sol. Using conservation of momentum along horizontal direction



mu sin 53° = $-mv_1$ sin 53° + Mv_2

$$\frac{4}{5}u = -v_1\frac{4}{5} + 4v_2$$

$5v_2 - v_1 = u$ …(i)

$$\frac{v_2\cos37° + v_1}{u} = e = 1$$

$0.8\,v_2 + v_1 = u$ …(ii)

Solving (i) and (ii), we get

$5.8\,v_2 = 2u$

$$\Rightarrow v_2 = \frac{2u}{5.8} = 2 \text{ m/s}$$

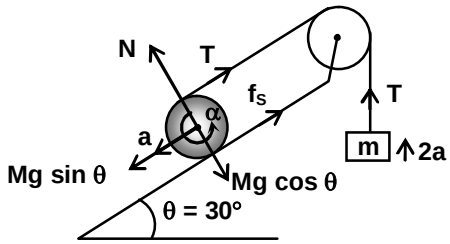
$\therefore v_1 = 5v_2 - u = 5 \times 2 - 5.8 = 4.2$ m/s

$$\int N dt = m(v_1 + u) = 10 \text{ N} - \text{s}$$

$$\int T dt = \int N\cos53°dt = \frac{3}{5}\int N dt = \frac{3}{5}\times10 = 6 \text{ N-s}$$

11. A, C, D

Sol. $Mg\sin\theta - T - f_S = Ma$ …(i)



$f_S R - TR = \frac{MR^2}{2}\alpha$

$f_s - T = \frac{Ma}{2}$ …(ii)

$T - mg = 2ma$

$2T - 2mg = 4ma$ …(iii)

Adding (i), (ii) and (iii), we get

$Mg\sin\theta - 2mg = \left(\frac{3M}{2} + 4m\right)a$

$\Rightarrow a = \left(\frac{M\sin\theta - 2m}{\frac{3M}{2} + 4m}\right)g = \frac{20}{16} = \frac{5}{4}$ m/s^2

$\therefore$ T = m(g + 2a) = 1(10 + 5/2) = 12.5 N

$\therefore f_S = T + \frac{Ma}{2} = 12.5 + \frac{8}{2}\times\frac{5}{4}$

$\Rightarrow f_S$ = 17.5 N

12. A, B, C, D

Sol. Use the definition of moment of Inertia.

SECTION – C

13. 00005.00

Sol. Apply work energy theorem

$mg - \frac{k}{2}\left(2\sqrt{2} - 2\right)^2 = 0$ => k=5

14. 00005.00

Sol. Draw the F.B.D. of the sphere at the top.

15. 00009.00

Sol. Let f_1 be the acceleration upwards of the 4 kg and therefore of the 1 kg. pulley downwards. Let f_2 and f_3 be the accelerations of the 2 and 3 kg (both measured downwards). Let T, T_1 be the tensions of the upper and lower strings. Then

$T - 4g = 4f_1$

$2T_1 - T + g = 1.f_1$

$2g - T_1 = 2f_2$

$3g - T_1 = 3f_3$

$f_2 + f_3 = 2f_1$

From the above equations => $f_1 = \frac{9g}{49}$

16. 00002.00

Sol. $v_1 \cos\theta = v_2 \sin\theta$

$v_1 = v_2 \tan\theta$

$v_1 = v_2\sqrt{3}$ (given $\theta = 60°$)

Now using conservation of energy

$$mga(1-\cos\theta) = \frac{1}{2}mv_1^2 + \frac{1}{2}Mv_2^2$$

$\Rightarrow \quad 2ga = 2v_1^2 + 4v_2^2$

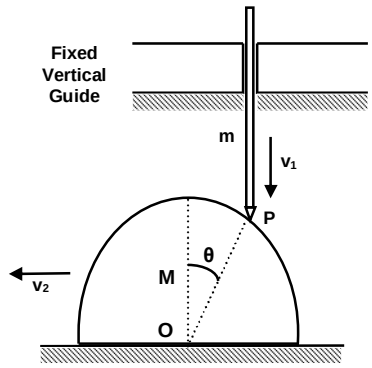
$40 = 6v_2^2 + 4v_2^2$

$\Rightarrow \quad 10v_2^2 = 40$

$\Rightarrow \quad v_2 = 2$ m/s

$\therefore \quad v_1 = 2\sqrt{3}$ m/s

$$\omega = \frac{v_1\sin\theta + v_2\cos\theta}{a} = 2 \text{ rad/s}$$



17. 00002.50

Sol. $v_0 = \omega h \sin\theta$

$$\Rightarrow \omega = \frac{v_0}{h\sin\theta} = \frac{2v_0}{h} = 2.50 \text{ rad/s}$$

18. 00004.50

Sol. When the ball A leaves the horizontal surface

$T = mg$ ….(i)

$$T + mg = \frac{m(2v)^2}{\ell}$$

$$2mg = \frac{4mv^2}{\ell}$$

$$v = \sqrt{\frac{g\ell}{2}}$$ …(ii)

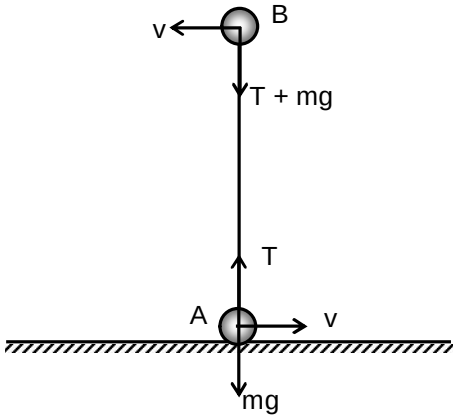
Now, using Conservation of Energy

$$\frac{1}{2}mu^2 = 2\times\frac{1}{2}mv^2 + mg\ell$$

$u^2 = 2v^2 + 2g\ell$

$u^2 = g\ell + 2g\ell$

$\Rightarrow u_{min} = \sqrt{3g\ell} = \sqrt{3\times10\times0.675} = 4.50$ m/s



Chemistry

PART – II

SECTION – A

19. A

Sol. $\frac{1}{\lambda} = R_H\left[\frac{1}{1^2} - \frac{1}{2^2}\right] = R_H\left[\frac{3}{4}\right]$

20. A

Sol. $BrO_3^- + I^- \longrightarrow I_2 + Br^-$

$I_2 + S_2O_3^{-2} \longrightarrow I^- + S_4O_6^{-2}$

Meq. of $KBrO_3$ = Meq. of $Na_2S_2O_3$

$$\frac{0.167 \times 1000}{\frac{167}{6}} = 75 \times N \times 1$$

$6 = 75\ N$

$$N = \frac{6}{75} = \frac{2}{25} = 0.08$$

21. B

Sol.

	N_2	+ $3H_2$ $\rightleftharpoons$	$2NH_3$
$t = 0$	1	3	0
$t = t$	$1 - 0.5$	$3 - 1.5$	2×0.5

$$P_{NH_3} = \frac{1}{3} \times 21 = 7 \text{ atm}$$

22. D

Sol. $Be(OH)_2 + NaOH \longrightarrow NaBeO_2$

23. B, D

Sol. (A) m = 2, $\ell = 2$, n = 3

(B) Ne = $1s^2\ 2s^2\ 2p^6$

(C) 3d = 3p

(D) $E \propto \frac{1}{n^2}$

n = 3

Third shell in hydrogen atom 3s = 3p = 3d

Ans. 9 orbitals.

24. A, B, C

Sol. It means electron would move into first excited state.

$$E_2 - E_1 = 13.6Z^2\left[\frac{1}{1^2} - \frac{1}{2^2}\right] = 10.2Z^2$$

Z may be 1, 2, 3……..

25. A, B, D

26. A, C, D

27. A, B, C, D

28. A, B, C, D

29. A, B

30. A, C

SECTION – C

31. 00006.00

Sol. Number of atoms in unit cell = 1 + 2 = 3

$$15 = \text{density} = \frac{3 \times \text{at.wt.}}{5 \times 10^{-24} \times 6 \times 10^{23}}$$

$$\text{At. Wt.} = 5 \times 5 \times 6 \times 10^{-1}$$

$$= 15$$

15 gram has 6×10^{23} atoms

$$150 \text{ gram } " \quad " = \frac{150}{15} \times 6 \times 10^{23}$$

$$= 60 \times 10^{23}$$

$$= 6 \times 10^{24}$$

$$x = 6$$

32. 00003.19

Sol. $NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$

$$2P \qquad P$$

$$K_p = 4P^3$$

$NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$

$2P - 0.5P \quad P + 1 - 0.25\,P$

$1.5P \qquad 0.75P + 1$

$$K_P = (1.5P)^2(0.75P + 1) = 4P^3$$

$$2.25(0.75P + 1) = 4P$$

$$\frac{9}{4}\left(\frac{3}{4}P + 1\right) = 4P$$

$$\frac{27P+36}{16}=4P$$

$$27P+36=64P$$

$$36=37P, \qquad P=\frac{36}{37}$$

$$P_T=1.5P+0.75P+1$$

$$=2.25P+1$$

$$=2.25\times\frac{36}{37}+1=3.19$$

33. 00000.50

Sol. $M\propto\frac{1}{x}$

$$3C_2H_2\longrightarrow C_6H_6$$

$$1 \qquad\qquad 0$$

$$1-\alpha \qquad\qquad \frac{\alpha}{3}$$

$$\frac{M_1}{M_2}=\frac{n_2}{n_1}$$

$$\frac{26}{39}=1-\frac{2\alpha}{3}$$

$$\frac{2\alpha}{3}=\frac{1}{3} \qquad \alpha=\frac{1}{2}$$

34. 00002.60

Sol. $A(s)\rightleftharpoons B(s)+2C(g)$

$$K_P=9=(P_C)^2$$

$$P_C=3$$

$$A(s)\rightleftharpoons B(s)+2C(g)$$

$$1 \quad a \text{ mol}$$

$$a-1\times 2$$

PV = nRT

$$3\times 10=(a-2)0.0821\times 600$$

$$(a-2)=\frac{3\times 10}{0.0821\times 600}=\frac{1}{0.0821\times 20}$$

a = 2.60

35. 00006.50

Sol. $pH=\frac{5+8}{2}=6.5$

$PO_4^{-3} + H^+ \longrightarrow HPO_4^{-2}$

5 mm 10 mm 0

0 5 mm 5 mm

$HPO_4^{-2} + H^+ \longrightarrow H_2PO_4^-$

5 mm 5 mm 0

0 0 5 mm

$H_2PO_4^-$ is a amphoteric ion whose, pH = $\frac{1}{2}\left(pK_{a_1} + pK_{a_2}\right) = \frac{1}{2}(5+8) = 6.5$

36. 00003.00

Sol. $13.6Z^2\left[\frac{1}{1^2} - \frac{1}{2^2}\right] = 13.6 \times 3^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

$Z^2 \times \frac{3}{4} = 9\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

$\frac{Z^2}{12} = \frac{1}{n_1^2} - \frac{1}{n_2^2}$

$Z = 1, n_1 = 3, n_2 = 6$

Hence $= \frac{2n_1 + 3n_2 + 6Z}{10} = 3$

Mathematics

PART – III

SECTION – A

37. A

Sol. Let $I(t)=\int_0^1 \frac{\sin(\ln x^t)}{\ln x}dx$

$I'(t)=\int_0^1 \frac{\ln x}{\ln x}\cos(t\ln x)dx = \int_0^1 \text{Re}\left(e^{it\ln|x|}\right)dx$

$I'(t)=\text{Re}\left(\frac{1-it}{1+t^2}\right)=\frac{1}{1+t^2}$

$\int_0^1 I'(t)dt=\frac{\pi}{4}$

38. A

Sol. $\int_0^1 \left[f^2(x^2)-2xf(x^2)+x\right]dx=0$

$\Rightarrow \int_0^1 \left(f(x^2)-x\right)^2 dx=0$

So, $f(x^2)=x$

i.e., $f(x)=\sqrt{x}$

39. D

Sol. $(f^3+g^2)f'+f^2gg'=0$

$\Rightarrow \left(\frac{1}{4}f^4+\frac{1}{2}f^2g^2\right)'=0$

$\Rightarrow f^4+2f^2g^2=c$

40. A

Sol. $y^4+xy-x^4-x^3y^3\geq 0$

$(y^3+x)(y-x^3)\geq 0$

Thus the two curves partition the circle $x^2+y^2\leq 2$ into 4 regions of equal area and the inequality is satisfied by the two regions the cover the y-axis. So, the area of the set is half the area of the circle.

41. A, B

Sol. By scaling and translating, we may assume that the two inflection points are located at x = –1 and x = 1

So, $f(x)=(x^4-6x^2+5)+bx+c$ where $2b=f(1)-f(-1)$; $2c=f(1)+f(-1)$ the line passing through inflection points is $g(x)=bx+c$

So, other intersection points = $\pm\sqrt{5}$

So, $\left|\int_{-\sqrt{5}}^{-1}(x^4-6x^2+5)dx\right| = \left|\int_{1}^{\sqrt{5}}(x^4-6x^2+5)dx\right| = \frac{16}{5}$ and $\int_{-1}^{1}(x^4-6x^2+5)dx = \frac{32}{5}$

42. A, B, C, D

Sol. Let $M = 3x^2 + 2e^y$, $N = 2xe^y + 3y^2$ solution of the differential equation is
$\int(3x^2+2e^x)dx + \int(3y^2)dy = c$
$\Rightarrow x^3 + 2xe^y + y^3 = c$

43. A, B, C, D

Sol. Let $F(t) = \left[\int_0^t f(x)dx\right]^2 - \int_0^t (f(x))^3 dx \ \ \forall\ t \in [0, 1]$

$F'(t) = f(t)\left[2\int_0^t f(x)dx - f^2(t)\right]$

Let $G(t) = 2\int_0^t f(x)dx - f^2(t)$

$G'(t) = 2f(t)(1-f'(t))$

So, $F'(t) \geq 0\ \forall\ t \in [0, 1]$

So, if $1 \geq t \geq 0$

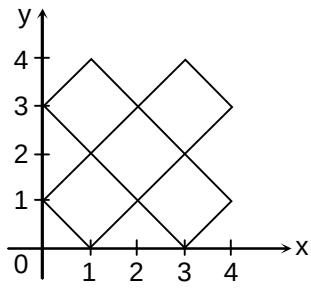
$F(1) \geq F(t) \geq F(0)$

$\left[\int_0^1 f(x)dx\right]^2 - \int_0^1 (f(x))^3 dx \geq 0$

If $f(x) = x$, $x \in [0, 1]$, then equality holds and in many more such example

44. A, C

Sol. Draw the graph in first quadrant



45. A, B, D

Sol. $f(x) = |\sin x| \cos^{-1}(\cos x)$
Points of non-differentiability $x = (2n + 1)\pi$
LHD = $-\pi$, RHD = 2π
$f(x) = [x]|\sin \pi x|$. Points of non-differentiability x = integers
LHD = $(1 - k)\pi$, RHD = $k\pi$
$f(x) = [x]\sin^2(\pi x)$ differentiable $\forall\ x \in R$
$f(x) = |x - 1|\{x\}$: points of non-differentiability x = 1

46. A, D

Sol. Clearly f(x) is continuous on [r, 5r] and not differentiable at x = –r, 5r
Because there is vertical tangent at x = –r, 5r

47. A, B

Sol. $\int \frac{x^4+1}{x^6+1}dx = \int \frac{x^4-x^2+1}{x^6+1}dx + \int \frac{x^2}{x^6+1}dx$

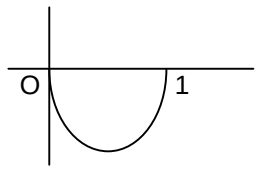
$= \int \frac{(x^4-x^2+1)}{(x^2+1)(x^4-x^2+1)}dx + \int \frac{x^2}{(x^3)^2+1}dx$

$= \tan^{-1}x + \frac{1}{3}\tan^{-1}x^3$

$= \frac{1}{3}\tan^{-1}\left(\frac{3x-3x^5}{1-3x^2-3x^4+x^6}\right)+c$

48. A, B, C

Sol. $(f(x)e^x)'' \geq 0$
So, $f(x)\, e^x$ is convex
So, $f(x)\, e^x \leq \max [f(0), f(1)] = 0$
$\therefore f(x) \leq 0 \ \forall \ x \in [0, 1]$



SECTION – C

49. 00002.00

Sol. Let $a = 2^x$, $b = -3^{x-1}$ and $c = -1$
So, $2^x = 3^{x-1}$
$\Rightarrow 3^{x-1} - 2^{x-1} = 2^{x-1} - 1$
Let $f(t) = t^{x-1}$
So, $f'(t_1) = f'(t_2)$ from LMVT on (1, 2) and (2, 3) and $t_1 \in (1, 2)$, $t_2 \in (2, 3)$
$(x-1)t_1^{x-2} = (x-1)t_2^{x-2}$
Clearly, there are only 2 solutions x = 1 and x = 2

50. 00000.20

Sol. $f(x) = \frac{\left(x+\frac{1}{x}-1\right)^3}{x^3+\frac{1}{x^3}-1}$

Let $x+\frac{1}{x} = t$

$g(t) = 1+\frac{6t-3t^2}{t^3-3t-1}$

After g′(t), we get CP: $t = 1, 1, 1\pm\sqrt{3}$

So, $g\left(1+\sqrt{3}\right) = \frac{\sqrt{3}}{2+\sqrt{3}}$

51. 00008.00

Sol. $f(x)=\left(x^2+\frac{a}{2}x\right)^2+\left(1+\frac{b}{2}x\right)^2+\frac{1}{4}(8-a^2-b^2)x^2\geq 0 \ \ \forall\, x\in R$

f(x) is strictly positive unless $a^2 + b^2 \geq 8$

52. 00007.00

Sol. $1+\frac{1}{2}+\frac{1}{3}+.....+\frac{1}{n^7}>\int_1^{n^7}\frac{dx}{x}>\frac{1}{2}+\frac{1}{3}+.....+\frac{1}{n^7}$

$\ln^7-1>1+\frac{1}{2}+\frac{1}{3}+.....+\frac{1}{n^3}>\ln n^7$

So, $\lim\limits_{n\to\infty}\frac{1+\frac{1}{2}+\frac{1}{3}+.....+\frac{1}{n^7}}{\ln n}=7$

53. 00001.00

Sol. $\because\ \frac{dx}{dy}=\frac{1}{\frac{dy}{dx}}$

$\frac{d^2x}{dy^2}=\frac{d}{dy}\left(\frac{1}{\frac{dy}{dx}}\right)=\frac{d}{dx}\left(\frac{1}{\frac{dy}{dx}}\right)\cdot\frac{dx}{dy}=\frac{-1}{\left(\frac{dy}{dx}\right)^2}\cdot\frac{d^2y}{dx^2}\cdot\frac{dx}{dy}=\frac{-1}{\left(\frac{dy}{dx}\right)^3}\cdot\frac{d^2y}{dx^2}$

From the given is $\frac{d^2y}{dx^2}\left(1-\frac{1}{\left(\frac{dy}{dx}\right)^3}\right)=0$

$\Rightarrow \frac{d^2y}{dx^2}=0$ or $\frac{dy}{dx}=1 \Rightarrow y = ax + b : a \neq 0$

54. 00000.00

Sol. Assume that such a function exists

$\because$ f′(x) = f(f(x)) > 0

$\Rightarrow$ f ↑

f′(x) = f(f(x)) > f(0) $\forall$ x $\in$ R

$\int_x^0 f'(x)dx>\int_x^0 f(0)dx$; x < 0

$\Rightarrow$ f(x) < (x + 1)f(0)

Clearly for x ≤ –1, f(x) ≤ 0

Contradicting the hypothesis that f(x) > 0 $\forall$ x

No such function exist